No reasoning phase transition at N = 24

Finite-size scaling of deductive-deduction depth, spin_gnn / sudoku-dd

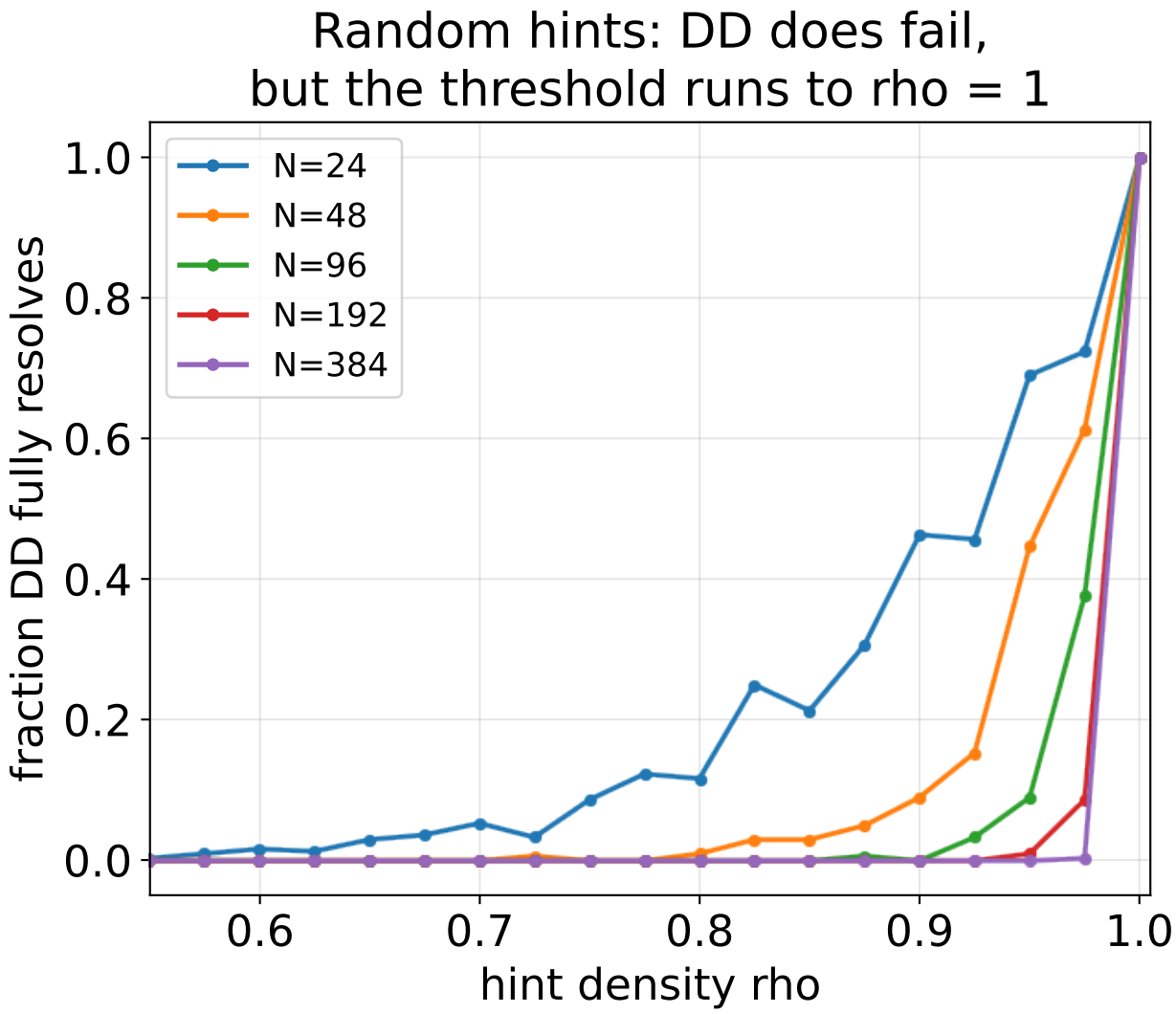
Headline: deduction depth T grows as a power of N at EVERY hint density. The exponent d lnT / d lnN drifts from +0.18 at rho=0.95 to +0.38 at rho=0.50 and never reaches zero, so there is no N-independent regime to sit outside of and no sharp critical density to sit at. N=24 is not near a transition -- it is far from the asymptotics.

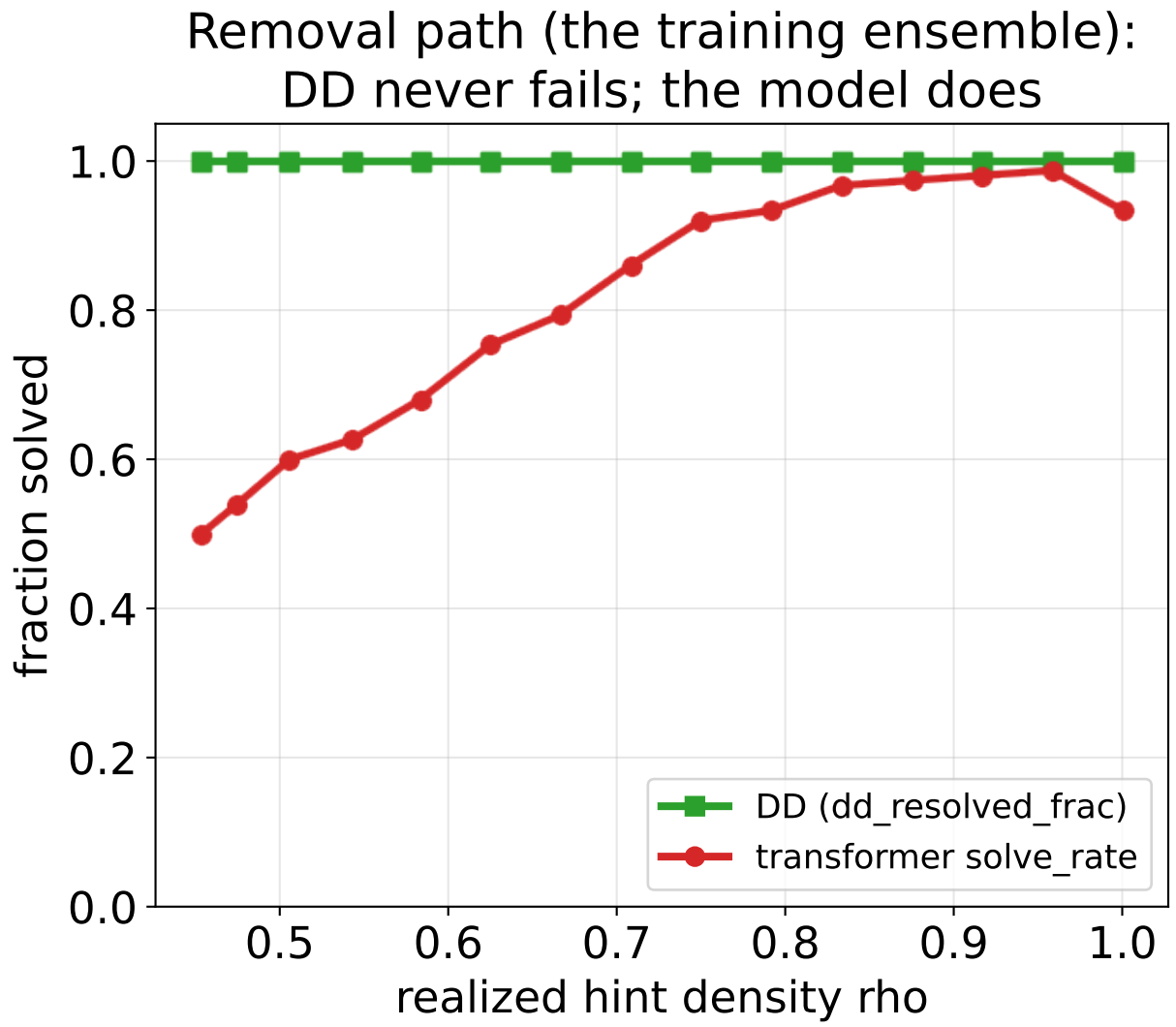
Two ensembles, neither with a solvability transition:
- random hints: threshold slides to rho=1 as N grows, unhinted nodes at the crossing stay O(1)
- removal path (what we train on): DD-solvable by construction, dd_resolved_frac = 1.0 everywhere

Source: jobs 48579545 / 48581589 / 48581590, 10-19. No new measurement, CPU-only re-analysis.

2026-10-24

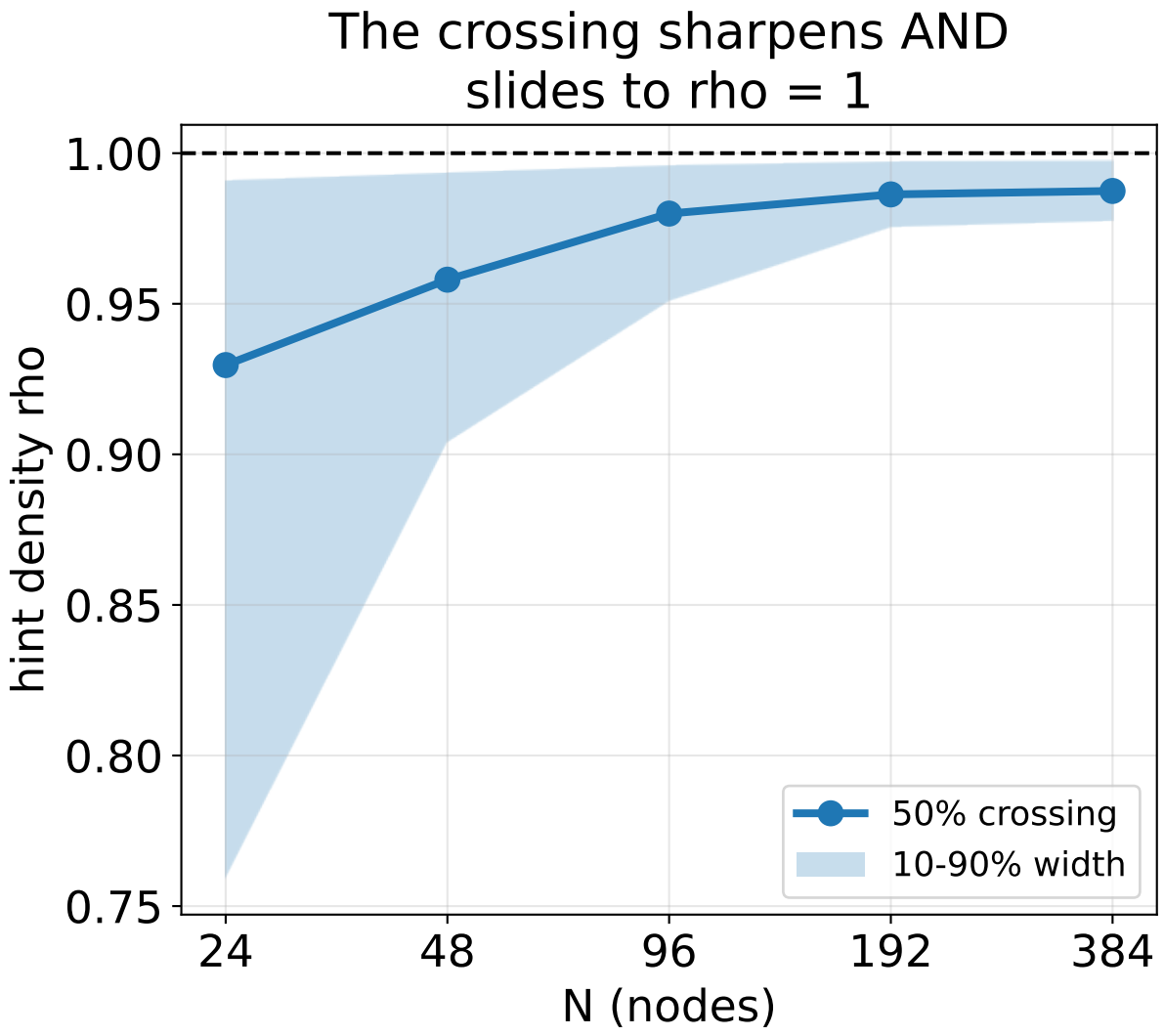
The 0.99 -> 0.44 curve is the model degrading, not DD transitioning

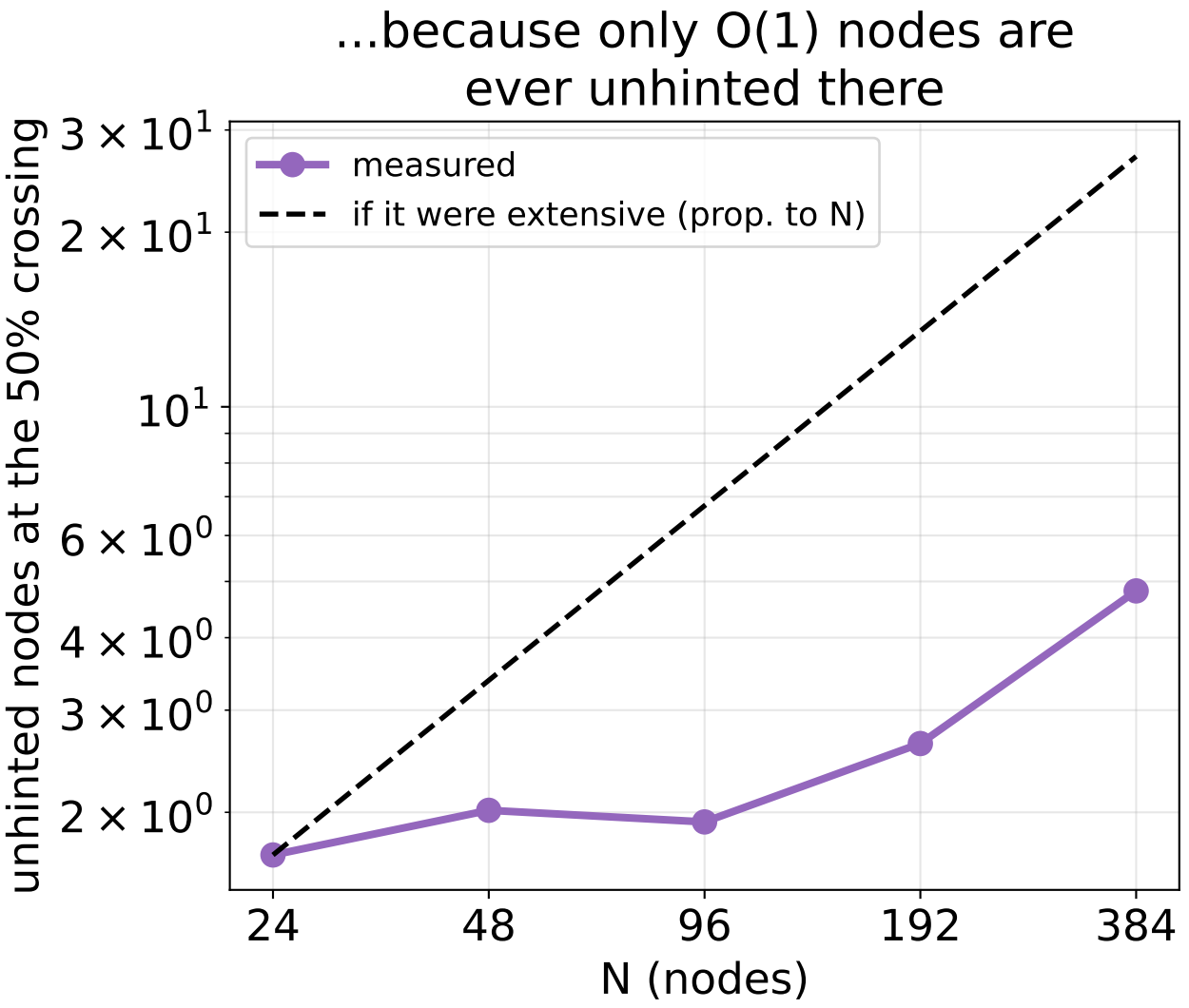




STATUS finding 1 said 'hint-density transition is real at N=24'. It is a transformer competence curve against a flat DD baseline.

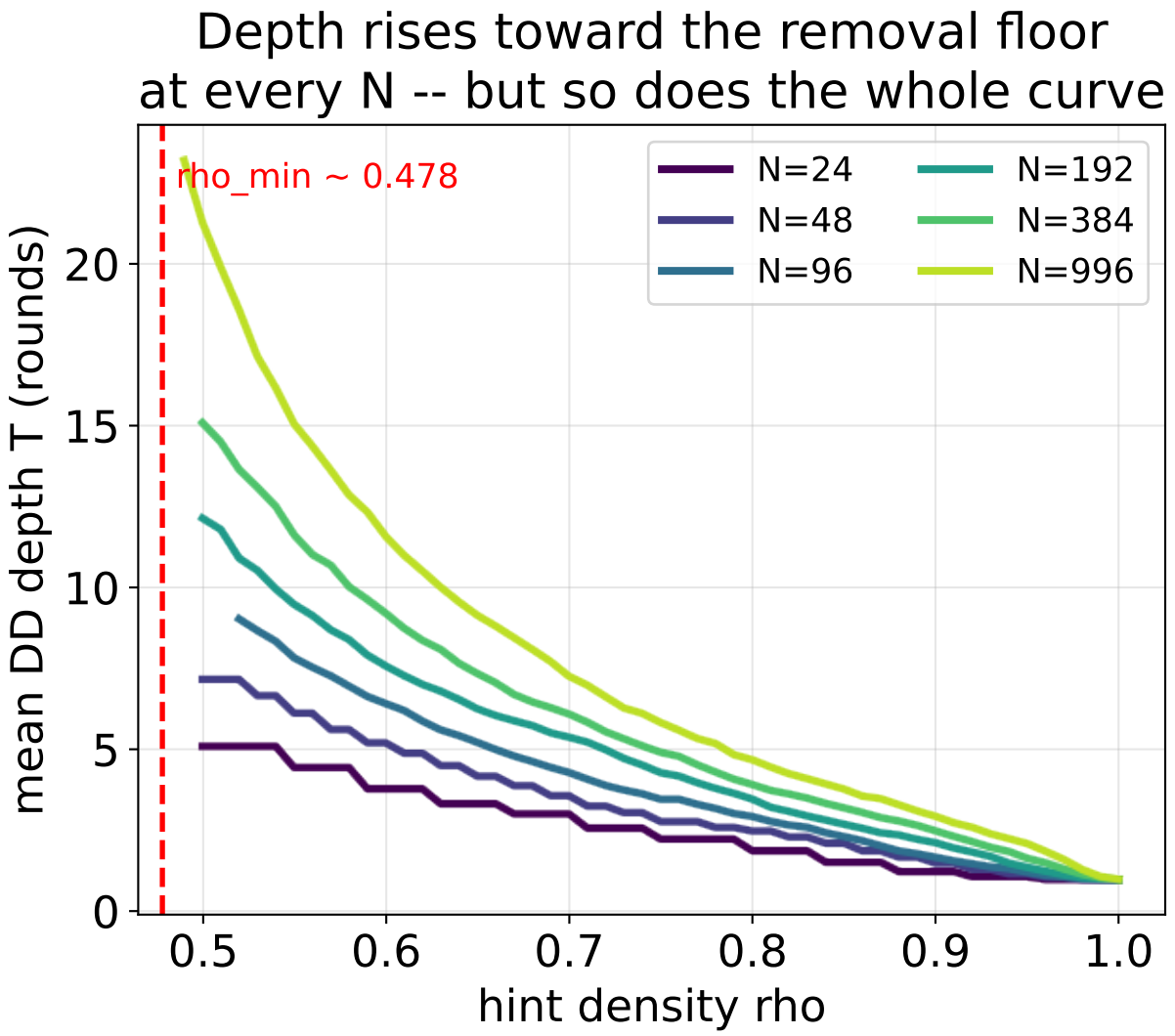
Random-hint ensemble: a threshold at zero unhinted density, not at finite rho

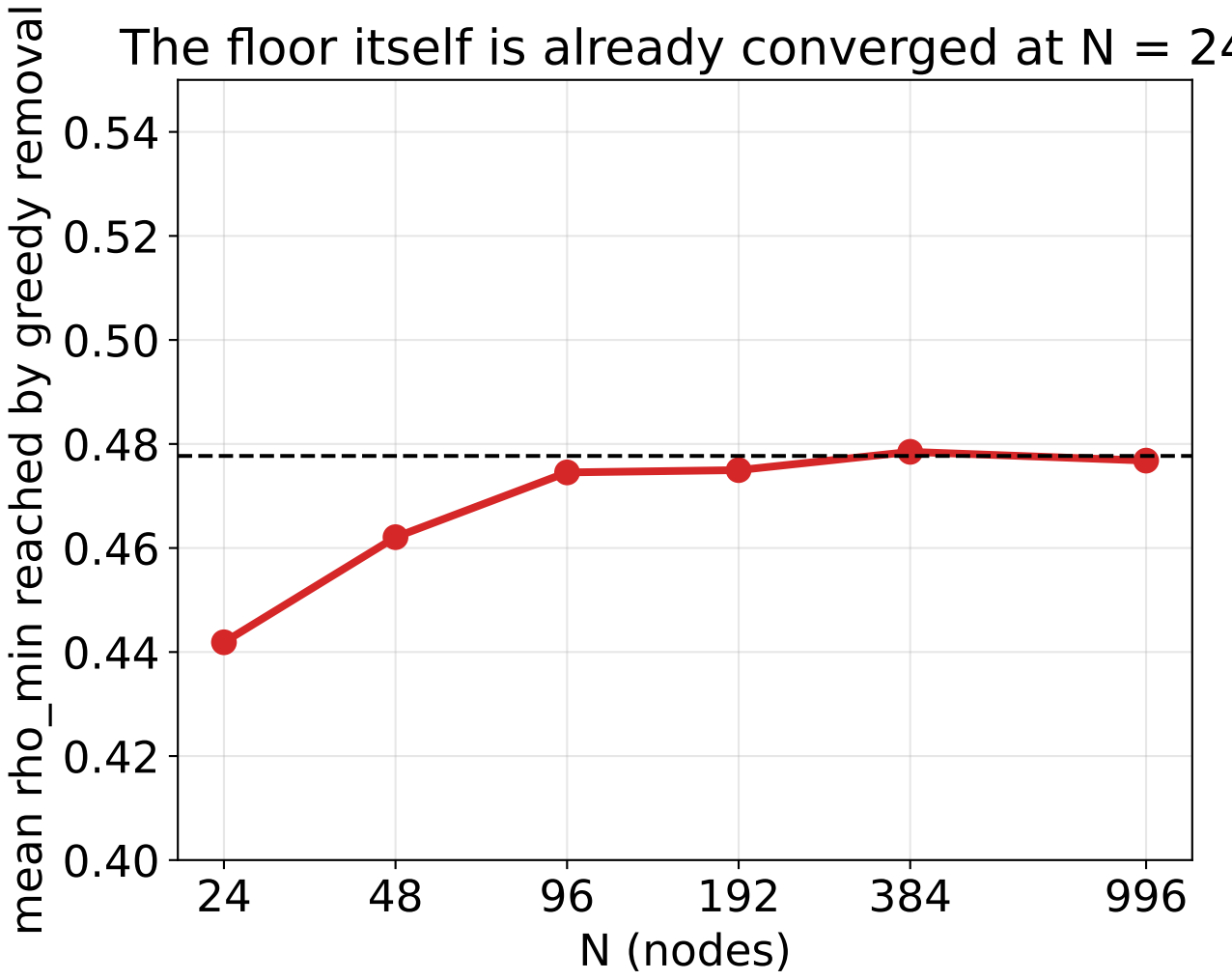




unhinted at crossing: 1.7, 2.0, 1.9, 2.6, 4.8 for N = 24, 48, 96, 192, 384. A finite-density transition would need this to grow 16x across this range; it grows 2.9x.

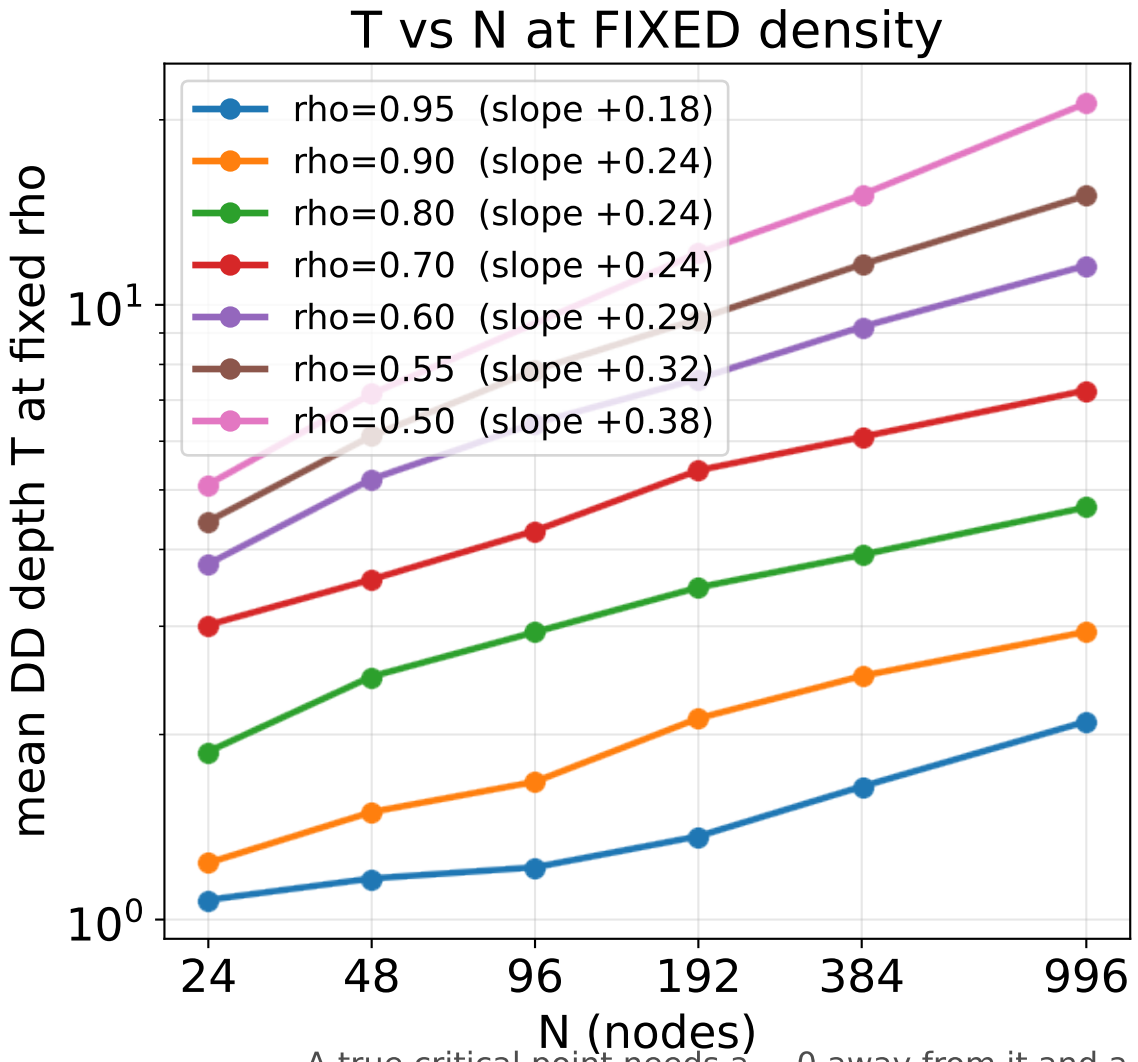
Removal-path ensemble: the floor location is N-independent, the depth is not

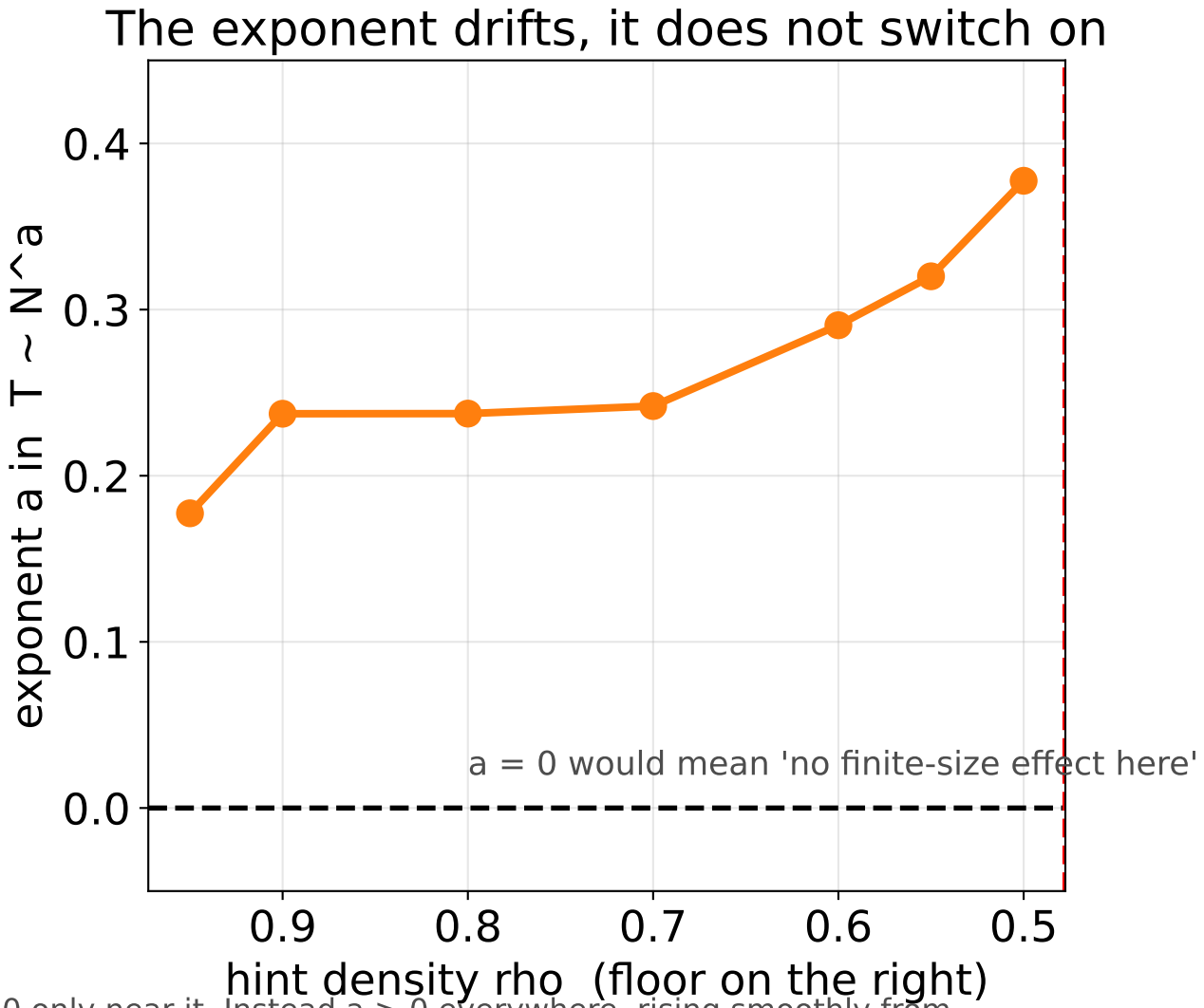




rho_min = 0.442, 0.462, 0.475, 0.475, 0.478, 0.477 for N = 24, 48, 96, 192, 384, 996. So going to larger N buys nothing on the density axis. Curves show only bins that at least 80% of removal paths reach.

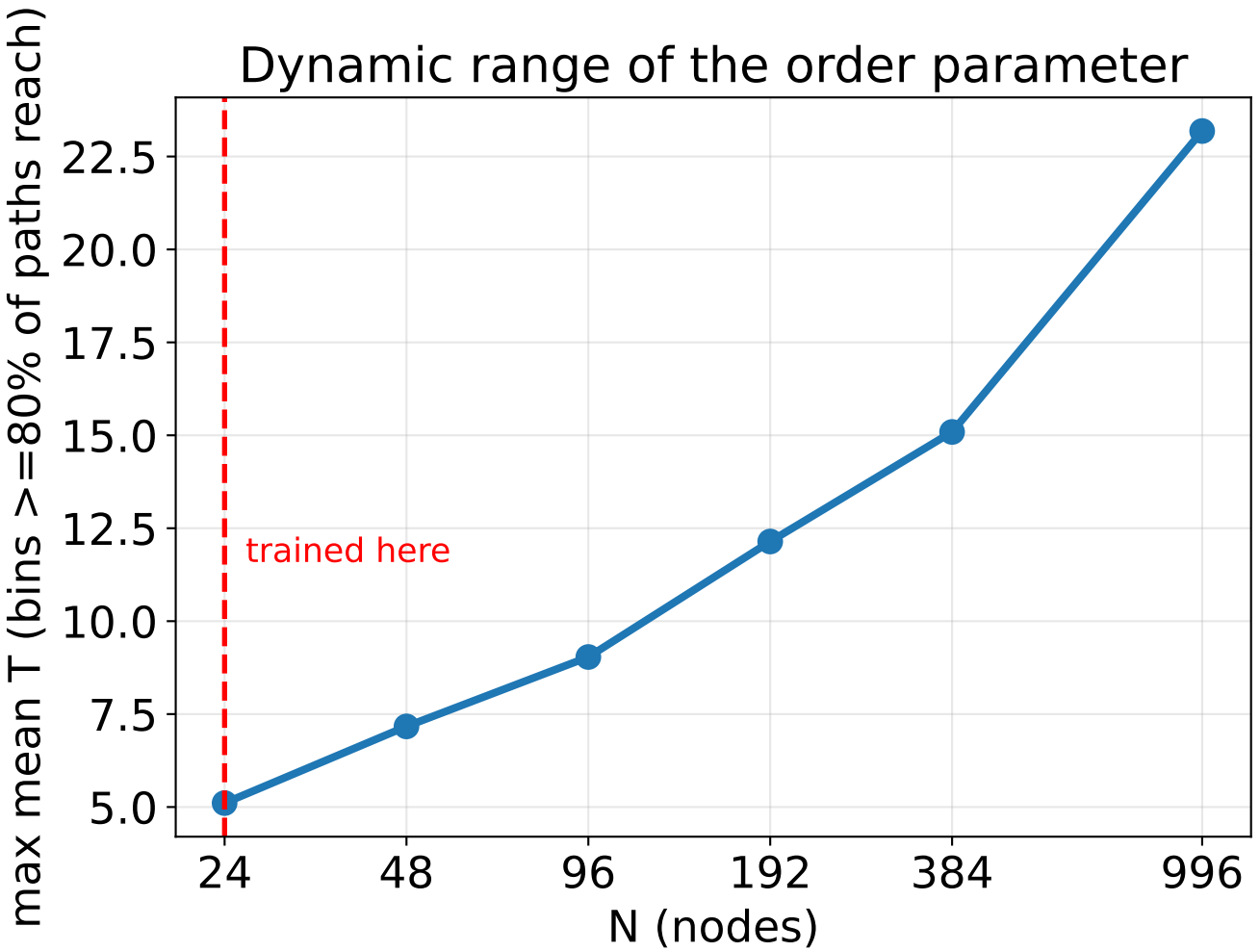
The scaling test: is rho_min a critical point, or just where greedy stops?





A true critical point needs a ~ 0 away from it and a > 0 only near it. Instead a > 0 everywhere, rising smoothly from 0.18 to 0.38. Verdict: no sharp critical density in this ensemble -- rho sets the exponent, it does not switch criticality on.

What N = 24 gives you, and what more would cost



Cost of raising N (quadratic attention)

N	tokens	cost	best recipe	max T
24	273	1x	5.5 GPU-h	5.1
48	537	4x	21.3 GPU-h	7.2
96	1065	15x	3.5 GPU-days	9.0
192	2121	60x	13.8 GPU-days	12.1

Doubling the order parameter's range costs ~16x in GPU time, and a range comparable to N=996 is out of reach entirely. 'max T' is the deepest bin at least 80% of removal paths reach; 5.5 GPU-h is the measured cost of the current best recipe (d0256 x 80000 x 300 ep, job 48912548). This is why the next move is a CPU sweep over (degree, n_colors), not a bigger N.

What is running, and what would change the picture

Job 49431733 (CPU-only, ~65 min): sweep DD over 20 (degree, n_colors) cells at N = 24 ... 996, both ensembles. Pre-registered:

```
(a)  rho_min rises as the propagation ratio d/(q-1) falls:
       rho_min(d=3,q=4) > 0.477 > rho_min(d=10,q=4)
(b)  some cell with d/(q-1) <= 1.2 reaches max T >= 22 at N=96,
       i.e. 1.5x the 14.9 that (5,4) gives there
(c)  falsifier: rho_min and max T flat in (d,q) to +-20%,
       meaning no cheap corner exists
(d)  scaling: T_at_rho90 flatter in N than T_at_rho55,
       if anything here is critical at all
(e)  control: unhinted-at-crossing stays O(1) in every random cell
```

Decision rule set in advance:

- If (b) holds, retrain at the winning (degree, n_colors) and keep N small -- cheap.
- If (c) holds instead, there is no cheap corner: the transition needs N ~ few hundred, which is GPU-weeks, and the honest move is to stop calling this a phase transition and study the depth-vs-competence curve on its own terms.

Either way, STATUS finding 1 gets rewritten: DD does not transition at N=24.